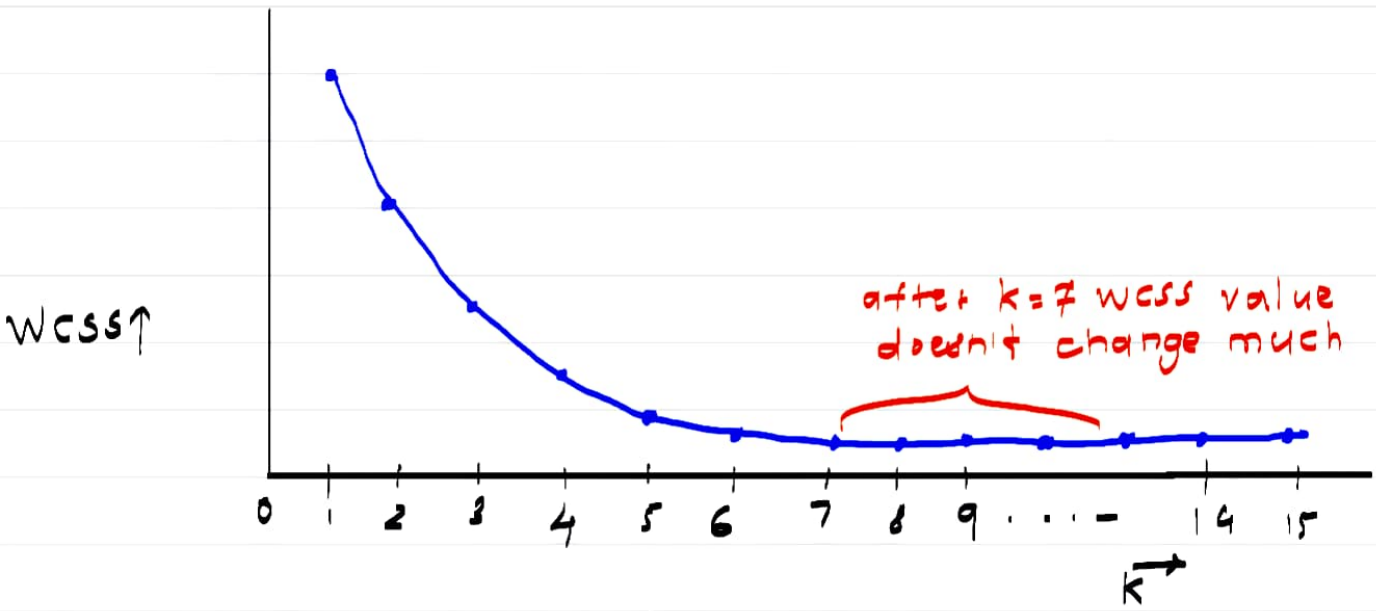
Machine Learning

20-04-26

- In KMeans clustering, model having lesser WCSS is the better model

- WCSS is also used to find optimal value of K.

K	WCSS
1	100
2	75
3	65
4	45
5	42
6	41.5
7	41.4
⋮	⋮
15	⋮



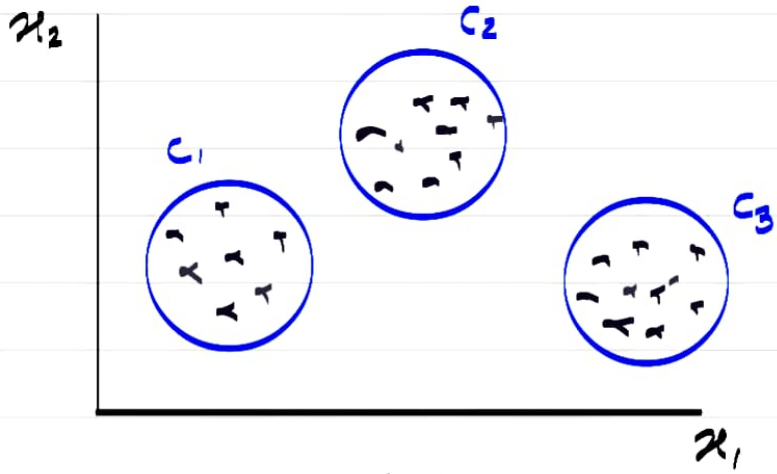
this is Elbow method for finding optimal value of K

After K=7, the WCSS value doesn't change much, hence K=7 is the optimal value in this case.

Silhouete Score

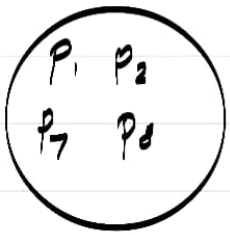
total 'n' datapoints $i = 1, 2, 3, \ldots n$

lets K=3 is choosen for KMeans clustering



datapoint	mean intra cluster distance	mean nearest cluster distance
P_1	a_1	b_1
P_2	a_2	b_2
P_3	a_3	b_3
P_4	a_4	b_4
⋮	⋮	⋮
P_n	a_n	b_n

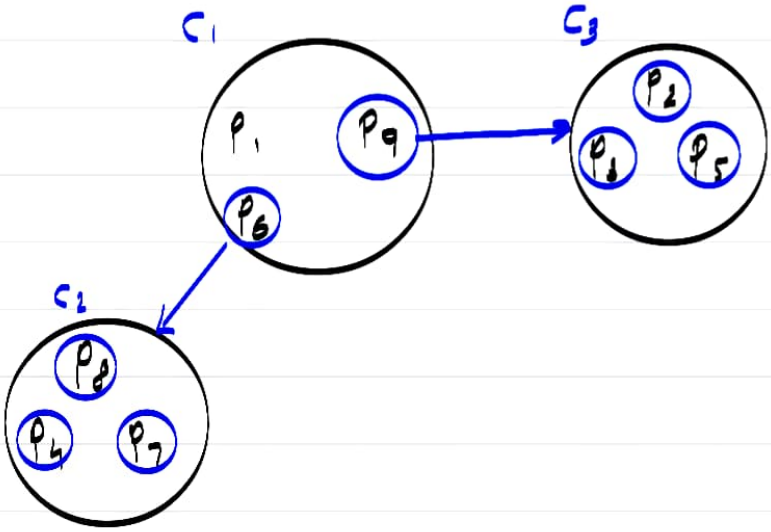
Mean Intra Cluster Distance: avg distance betn ith datapoint and all other datapoints within same cluster.



$$a_1 = \text{MICD for } P_1 = \frac{dist(P_1, P_2) + dist(P_1, P_7) + dist(P_1, P_8)}{3}$$

$$a_7 = \text{MICD for } P_7 = \frac{dist(P_7, P_1) + dist(P_7, P_2) + dist(P_7, P_8)}{3}$$

Mean Nearest Cluster Distance : avg distance betn ith datapoint and all datapoints from the nearest cluster



$dist(P_9, C_2) > dist(P_9, C_3)$

for P_9 nearest cluster is C_3

$$b_9 = MNCD \text{ for } P_9 = \frac{dist(P_9 P_2) + dist(P_9 P_3) + dist(P_9, P_5)}{3}$$

$dist(P_6, C_2) < dist(P_6, C_3)$

for P_6 nearest cluster is C_2

$$b_6 = MNCD \text{ for } P_6 = \frac{dist(P_6 P_4) + dist(P_6 P_8) + dist(P_6 P_7)}{3}$$

for ith datapoint

$$\boxed{\text{Silhouetee Score: } \frac{b_i - a_i}{\max(b_i, a_i)}}$$

p_1	a_1	b_1	s_1
p_2	a_2	b_2	s_2
p_3	a_3	b_3	s_3
$\vdots$	$\vdots$	$\vdots$	$\vdots$
p_n	a_n	b_n	s_n

avg silhouetee score $= \frac{1}{n} \sum_{i=1}^{n} s_i$

silhouetee score $\propto$ model performance

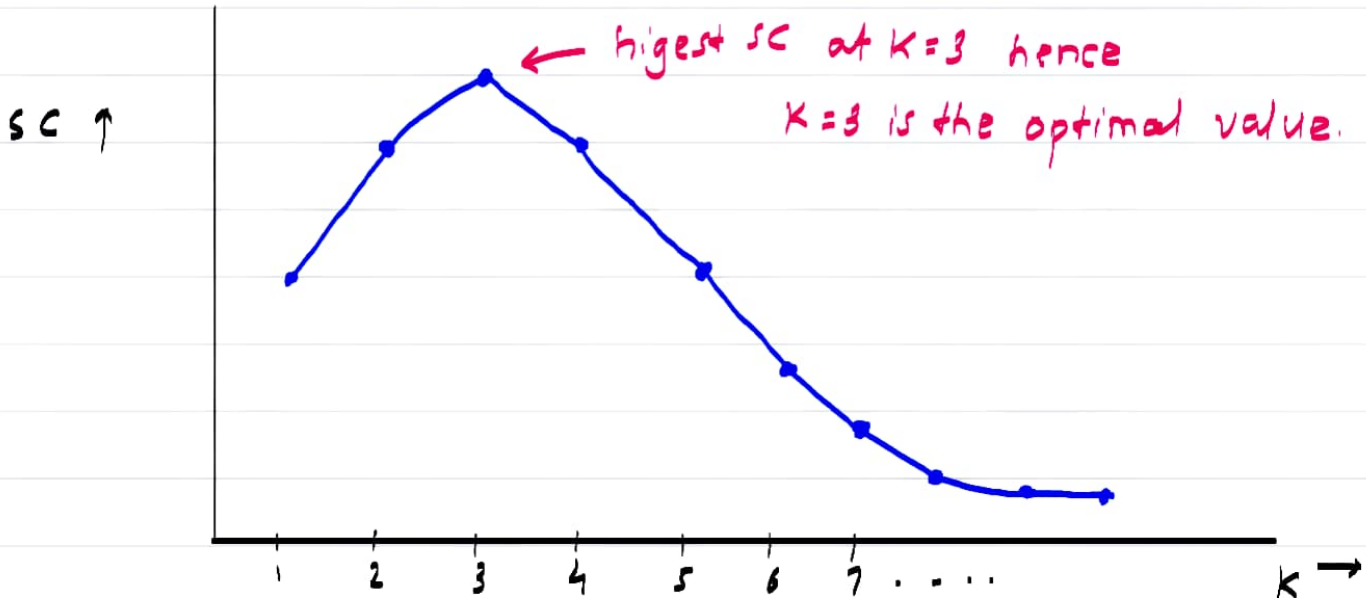
high SC $\Rightarrow$ better model

$-1 \leq$ silhouetee score ≤ 1

if for $K=3$ and $K=5$

$SC(K=3) > SC(K=5)$ then

$K=3$ is the better clustering model



How KMeans is related to RAG?

RAG - Retrieval Augmented Generation

ML book → Q.- What is silhoutee score?

10k pages PDF → chatgpt

5M tokens/words

all LLMs have a limited context window = 1M tokens

chatgpt can only read and store 1M tokens from book, rest 4M will be simply ignored.

Pages - P_1 P_2 P_3 .... P_{10k}

↓

embedding generation model

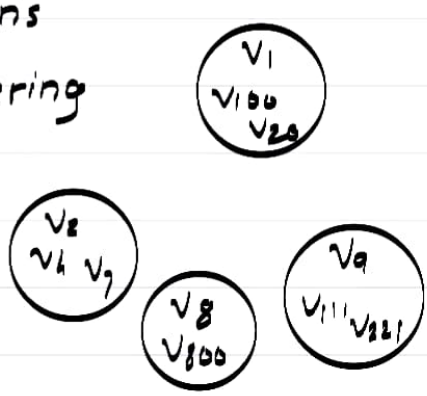
↓

V_1 V_2 V_3 V_4 .... V_{10k}

store all semantic information in given page

$V_1 = [0.4, 0.789, 0.123 \ldots\ldots 0.578]$

$V_1\ V_2\ V_3 \ldots V_n$ → vector store $\xrightarrow[\text{clustering}]{\text{KMeans}}$



Q.- what is silhoutee score?

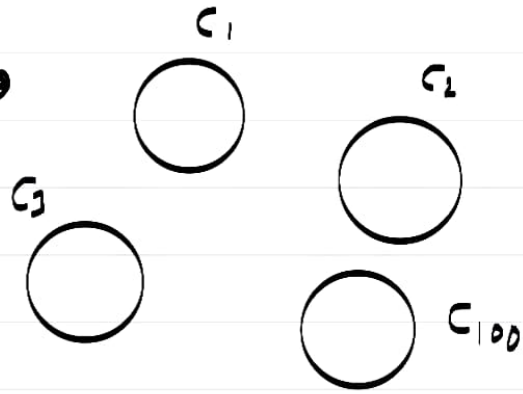
↓

embedding generation model

↓

$q = [-1.12, 0.72, -0.76, 0.51, 0.43 \ldots]$

$q \longrightarrow$ vector store $\longrightarrow$



$$\left.\begin{array}{l} dist(q, C_1) \\ dist(q, C_2) \\ dist(q, C_3) \\ dist(q, C_4) \\ \vdots \\ dist(q, C_{100}) \end{array}\right\} \text{nearest cluster for } q = C_8$$

↓

200 pages

$P_7, P_9, P_{1000} \ldots$

in the next step we find top N nearest pages for q in C_8

lets say for $N = 7$

find 7 nearest pages for q

$P_8 \quad P_{1000} \quad P_{111} \quad P_{221} \quad P_{78} \quad P_{99} \quad P_2$

↓

Q.- what is silhouetee score? $\longrightarrow$ Chatgpt $\longrightarrow$ <u>Ans</u>

Decision Tree Algorithm

- supervised ML
- used for regression and classification